Chapter 5, Problem 80P

Problem

Use *PSpice* or *MultiSim* to solve Prob. 5.70.

Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

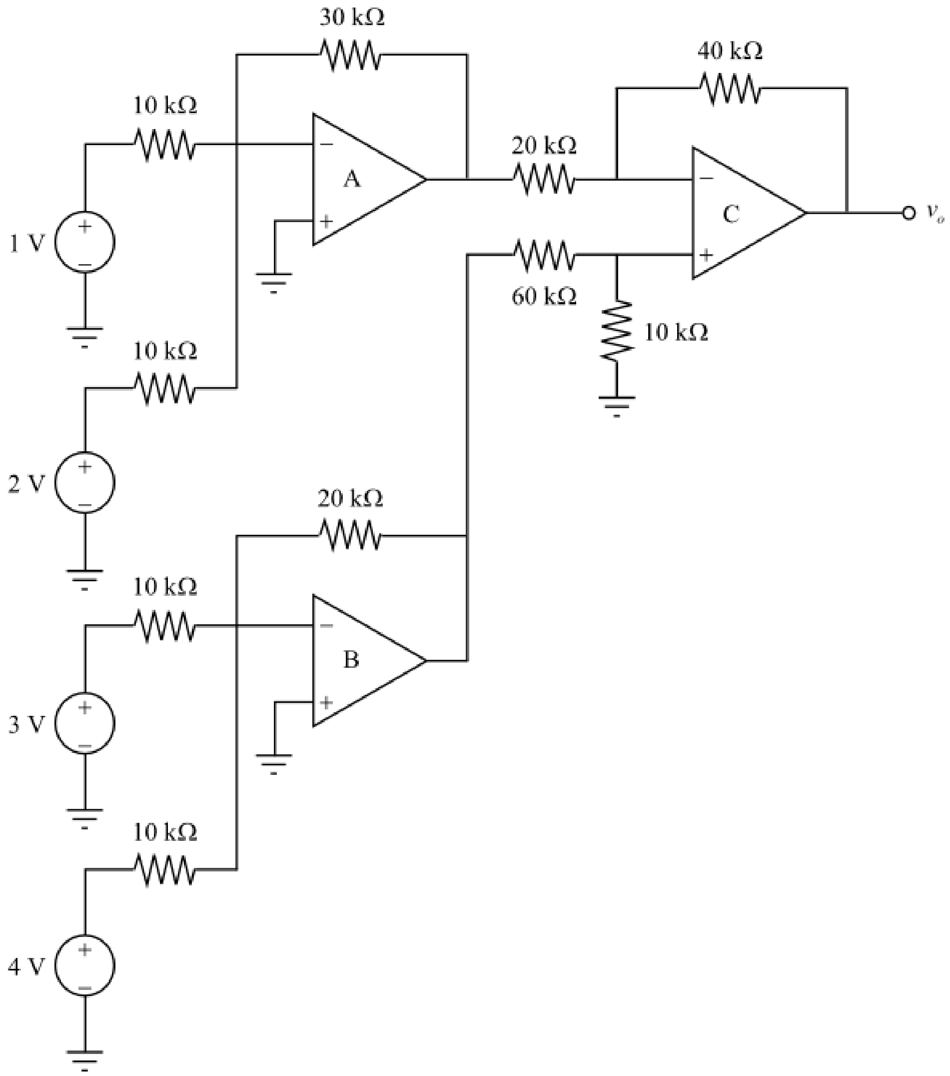

Figure 1

Step 2 of 6

The operational amplifiers A, and B in the circuit of Figure 1 are summing amplifiers.

The summing amplifier combines several inputs and produces an output as a weighted sum of the inputs.

Assume the output of operational amplifier A as v_1.

The summing amplifier A in the circuit has two inputs 1 V and 2 V.

The following is the output of the summing amplifier A.

$$v_1 = -\left[\frac{30\ \text{k}\Omega}{10\ \text{k}\Omega}(1\ \text{V}) + \frac{30\ \text{k}\Omega}{10\ \text{k}\Omega}(2\ \text{V})\right]$$

Simplify this expression to get the following.

$$\begin{aligned} v_1 &= -\left[(3)(1\ \text{V}) + (3)(2\ \text{V})\right] \\ &= -(3\ \text{V} + 6\ \text{V}) \\ &= -9\ \text{V} \end{aligned}$$

Step 3 of 6

Assume the output of operational amplifier B as v_2.

The summing amplifier A in the circuit has two inputs 3 V and 4 V.

The following is the output of the summing amplifier B.

$$v_2 = -\left[\frac{20\ \text{k}\Omega}{10\ \text{k}\Omega}(3\ \text{V}) + \frac{20\ \text{k}\Omega}{10\ \text{k}\Omega}(4\ \text{V})\right]$$

Simplify this expression to get the following.

$$\begin{aligned} v_2 &= -\left[(2)(3\ \text{V}) + (2)(4\ \text{V})\right] \\ &= -(6\ \text{V} + 8\ \text{V}) \\ &= -14\ \text{V} \end{aligned}$$

Step 4 of 6

The output of operational amplifier C is the output v_o.

The operational amplifier C in the circuit has two inputs from summing amplifiers A and B, that are -9 V and -14 V.

Redraw the circuit in Figure 1 as follows to calculate the output v_o.

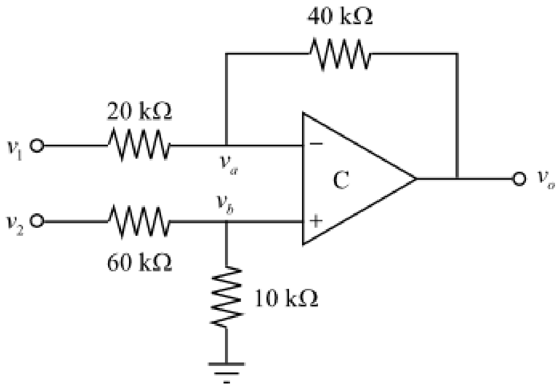

Figure 3

The current flow in to the input terminals of an ideal operational amplifier is zero.

Apply Kirchhoff's current law at node b to find the voltage, v_b.

$$\frac{v_2 - v_b}{60\text{ k}\Omega} = \frac{v_b}{10\text{ k}\Omega}$$
$$10\text{ k}\Omega(v_2 - v_b) = 60\text{ k}\Omega(v_b)$$
$$(10\text{ k}\Omega)v_2 = (60\text{ k}\Omega)v_b + (10\text{ k}\Omega)v_b$$
$$(10\text{ k}\Omega)v_2 = (70\text{ k}\Omega)v_b$$

Substitute -14 V for v_2.

$$(10\text{ k}\Omega)(-14\text{ V}) = (70\text{ k}\Omega)v_b$$

$$\begin{aligned} v_b &= \frac{-140\text{ kA}}{70\text{ k}\Omega} \\ &= -2\text{ V} \end{aligned}$$

Step 5 of 6

The voltage at the inverting and noninverting terminals of an ideal operational amplifier is same, since there is no current flow in to the input terminals of the operational amplifier. Therefore,

$$\begin{aligned} v_a &= v_b \\ &= -2\text{ V} \end{aligned}$$

Apply Kirchhoff's current law at node a to find the voltage, v_a.

$$\frac{v_1 - v_a}{20\text{ k}\Omega} = \frac{v_a - v_o}{40\text{ k}\Omega}$$
$$40\text{ k}\Omega(v_1 - v_a) = 20\text{ k}\Omega(v_a - v_o)$$
$$(40\text{ k}\Omega)v_1 + (20\text{ k}\Omega)v_o = (20\text{ k}\Omega)v_a + (40\text{ k}\Omega)v_a$$
$$(20\text{ k}\Omega)v_o = (60\text{ k}\Omega)v_a - (40\text{ k}\Omega)v_1$$

Substitute -2 V for v_a, and -9 V for v_1.

$$(20\text{ k}\Omega)v_o = (60\text{ k}\Omega)(-2\text{ V}) - (40\text{ k}\Omega)(-9\text{ V})$$
$$(20\text{ k}\Omega)v_o = -120\text{ kA} + 360\text{ kA}$$
$$(20\text{ k}\Omega)v_o = 240\text{ kA}$$

$$\begin{aligned} v_o &= \frac{240\text{ kA}}{20\text{ k}\Omega} \\ &= 12\text{ V} \end{aligned}$$

Thus, the output of the circuit, v_o is $\boxed{12\text{ V}}$.

Draw the circuit of Figure 1 in PSPICE to solve for the output voltage, v_o:

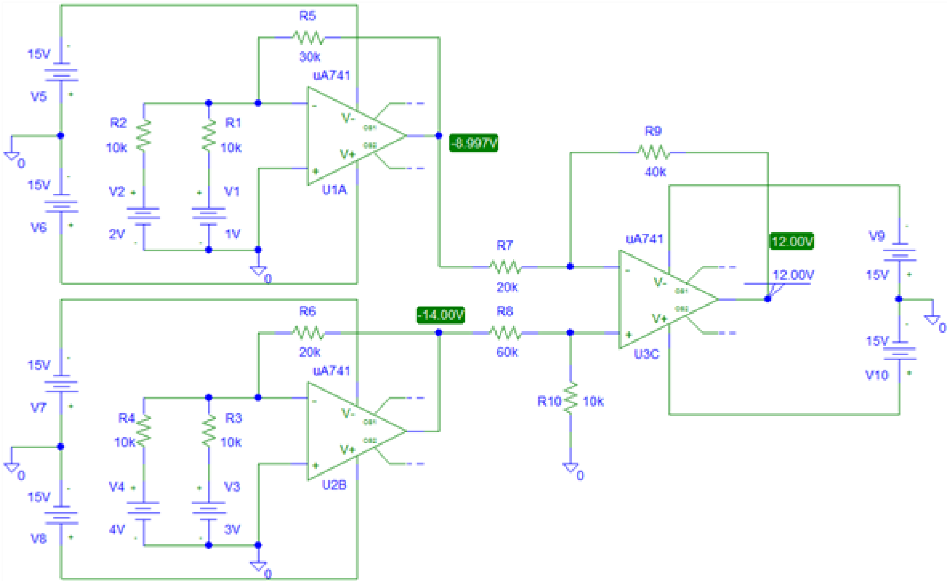

Figure 4